

SP-19 Phase 4 Extension: ABC, Hilbert 12th, Smooth Poincare

Scoped by: Keeper (May 13, 2026) Directive: Casey — “improve our ABC, Hilbert, and Smooth Poincare. Leave existing work as-is, use new work as exploration.” Status: SCOPED, awaiting team assignment

Investigation A: ABC Extension (Szpiro for all Heegner CM curves)

What we have

Toy 2169 (22/22): Szpiro ratio $\sigma = \log|\Delta|/\log(N) = 3/2 = N_c/\text{rank}$ for 49a1. EXACT.

What’s new (pre-scoping computation)

Szpiro ratio computed for ALL 9 Heegner CM curves ($h(-d) = 1$):

disc	d	N		Delta	
-3	3	$27 = 3^3$	$27 = 3^3$	1.000	no ($j=0$, extra auts)
-4	4	$32 = 2^5$	$32 = 2^5$	1.000	no ($j=1728$, extra auts)
-7	7	$49 = 7^2$	$343 = 7^3$	1.500	YES
-8	8	$256 = 2^8$	$512 = 2^9$	1.125	no (2 ramifies)
-11	11	$121 = 11^2$	$1331 = 11^3$	1.500	YES
-19	19	$361 = 19^2$	$6859 = 19^3$	1.500	YES
-43	43	$1849 = 43^2$	$79507 = 43^3$	1.500	YES
-67	67	$4489 = 67^2$	$300763 = 67^3$	1.500	YES
-163	163	$26569 = 163^2$	$4330747 = 163^3$	1.500	YES

Pattern: $\sigma = 3/2$ for exactly the 6 Heegner numbers where $N = d^2$ and $|\Delta| = d^3$ (all odd $d \geq 7$). The 3 exceptions ($d = 3, 4, 8$) have extra automorphisms ($j = 0$ or 1728) or even discriminant.

BST interpretation: $\sigma = N_c/\text{rank}$ is NOT special to 49a1. It holds for 6/9 Heegner CM curves. The exceptions are exactly the curves with extra symmetry ($\text{Aut}(E) > \mathbb{Z}/2\mathbb{Z}$) or bad behavior at 2.

Deliverables

Toy A1 (Elie): Szpiro table for all 9 Heegner CM curves. - Point-count a_p for each curve at small primes - Verify $N = d^2$ vs $N = d^k$ for each - Verify $|\Delta| = d^3$ for each - Compute σ , classify which are $3/2$ - Identify WHY $d=3,4,8$ differ (extra automorphisms explanation) - Check: are the 6 curves with $\sigma=3/2$ exactly those with $\text{Aut}(E) = \mathbb{Z}/2\mathbb{Z}$? - BST integer map for each curve’s invariants - Target: 20-25 tests

Toy A2 (Elie): Szpiro for NON-CM curves with small conductor. - Pick 10-15 curves from Cremona (conductors 11-100, non-CM) - Compute σ for each - Question: is σ always ≥ 1 ? Is $3/2$ a natural threshold? - Does the ABC bound $\sigma < 6 + \epsilon$ (Szpiro’s conjecture) hold with BST-structured margins? - Target: 15-20 tests

Toy A3 (Elie + Lyra): The Frey curve connection. - For an ABC triple (a, b, c) with $a+b=c$, the Frey curve is $y^2 = x(x-a)(x+b)$ - Conductor $N = \text{rad}(abc)$, discriminant $\Delta = (abc)^2/16$ - $\sigma = \log|\Delta|/\log(N) = 2\log(abc)/\log(\text{rad}(abc))$ - This is $2 \sum(v_p * \log(p)) / \sum(\log(p))$ — a weighted average - Question: does BST constrain which triples are “allowed”? - Specific: for the 49a1-related triple (g^2, g^2C_2, g^3) , verify $\sigma = N_c/\text{rank}$ directly - Target: 15-20 tests

Paper potential: “Szpiro’s Ratio and the BST Integers: Evidence from CM Curves” — short note showing $\sigma = N_c/\text{rank}$ for 6/9 Heegner curves, explaining the 3 exceptions. Venue: Experimental Mathematics or IMRN (short).

Honest scope: This does NOT prove ABC. It shows BST integers organize the Szpiro landscape for CM curves. The general ABC conjecture remains external.

Investigation B: Hilbert's 12th Extension (Stark Units)

What we have

Toy 2171 (25/25): Hilbert's 12th solved for $Q(\sqrt{-7})$ via CM of 49a1. All arithmetic in BST integers.

What's genuinely open

Hilbert's 12th for imaginary quadratic fields is SOLVED (Kronecker's Jugendtraum / CM theory). What's open:

1. Real quadratic fields — Stark's conjecture provides conjectural generators 2. Totally real fields — Shintani's work, partial results 3. Non-abelian extensions — wide open

The BST question

Does D_{IV}^5 encode Stark units for real quadratic fields? Specifically:

Stark's conjecture for $Q(\sqrt{d})$: There should exist a unit ϵ in the Hilbert class field such that $\log|\epsilon| = -L'(0, \chi_d)$. The conjecture gives explicit generators of abelian extensions.

BST test fields: $Q(\sqrt{5})$ and $Q(\sqrt{2})$ — these involve BST integers ($n_C = 5$, rank = 2).

Deliverables

Toy B1 (Lyra): Stark unit landscape for BST-adjacent real quadratic fields. - Compute $L'(0, \chi_d)$ for $d = 2, 3, 5, 6, 7$ (BST integers and neighbors) - Evaluate Stark units (known numerically for small d) - Check: are Stark units expressible in BST integers? - For $Q(\sqrt{5})$: fundamental unit $= (1+\sqrt{5})/2 = \phi$. Is $\log(\phi)$ BST-structured? - For $Q(\sqrt{7})$: fundamental unit $= 8+3\sqrt{7}$. Is $\log(8+3\sqrt{7})$ BST-related? - Compare: imaginary quadratic (SOLVED) vs real quadratic (CONJECTURAL) - Target: 20-25 tests

Toy B2 (Lyra + Elie): Hecke L-function evaluations on D_{IV}^5 . - The Eisenstein series on D_{IV}^5 at special points should encode L-values - For CM forms: $L(1, \chi_{-d}) = \pi/\sqrt{d} * (\text{class number formula})$ - For real quadratic: $L(1, \chi_d) = (2\log(\epsilon))/\sqrt{d} h(d)$ - Question: does the D_{IV}^5 Eisenstein machinery produce Stark units via its boundary values? - Specifically: evaluate $E(f, s, P_2)$ at $s = 1/2$ (functional equation center) — do Stark units appear? - Target: 15-20 tests

Toy B3 (Lyra): The 9 Heegner fields as a complete system. - All 9 imaginary quadratic fields with $h=1$ as a unified CM system - BST integer content of each: which j -invariants, conductors, discriminants are BST expressions? - The "BST Heegner numbers" $\{1, 2, 3, 7\}$ vs $\{11, 19, 43, 67, 163\}$ - Product: $1237 = 42 = C_{2g}$. Product of all 9? - Question: does the Gross-Zagier formula for Heegner points on 49a1 give BST integers? - Target: 20-25 tests

Paper potential: If Stark units for $Q(\sqrt{5})$ or $Q(\sqrt{7})$ are BST-structured, that's a genuine new result connecting D_{IV}^5 to explicit class field theory for real quadratic fields. Venue: Journal of Number Theory. If NOT structured, document the boundary honestly — where CM ends and the open frontier begins.

Honest scope: The imaginary quadratic case is classical. Real quadratic is conjectural (Stark). We're testing whether BST reaches into Stark's territory. High risk, high reward.

Investigation C: Smooth Poincare Dim 4 (Donaldson Connection)

What we have

- Toy 2168 (22/22): Gauss-Codazzi under-determined at $d=4$, excess = rank = 2, deficit = $1/n_C$
- Cal's GC-3 investigation: tangent-space argument (D_{IV}^5 inherits standard smooth structure)
- GC-6 dimension ladder: $d=4$ is the unique pathological row

The key insight from Cal (GC-3)

“Physics on D_{IV}^5 inherits R^4 as a tangent space, but the smooth structure inherited from the complex embedding is uniquely the standard one. The exotic R^4 s are smoothings that cannot arise as tangent spaces to a Hermitian complex manifold.”

This is the right framing: BST constrains R^4 from above (via the embedding), not from within.

What’s genuinely needed

Three sub-investigations, increasing difficulty:

C1: The $SU(2)$ connection (tractable) Donaldson invariants for 4-manifolds come from $SU(2)$ gauge theory. $SU(2)$ embeds in $SO(5,2)$: - $SU(2)$ subset $SO(4)$ subset $SO(5)$ subset $SO(5,2)$ - The moduli space of anti-self-dual $SU(2)$ connections on a 4-manifold M has dimension: $\dim M_{\{ASD\}} = 8k - 3(1+b^+)$ where k = instanton number, b^+ = positive second Betti - For S^4 : $b^+ = 0$, so $\dim = 8k - 3$ - Question: does the rank-2 excess from Toy 2168 correspond to the 2 parameters in the “smallest” non-trivial moduli space?

C2: Seiberg-Witten from D_{IV}^5 (harder) The SW invariants use Spin^c structures. D_{IV}^5 has a natural Spin^c structure (it’s Kahler). - The SW equations on a Kahler manifold reduce to vortex equations - D_{IV}^5 restricted to a 4-dimensional submanifold: what do SW invariants see? - Question: does the Bergman metric on D_{IV}^5 force trivial SW invariants on 4-dimensional slices? - If yes: BST predicts no exotic structure on the 4-dimensional physics it describes

C3: The intersection form (deepest) Freedman: topological 4-manifolds classified by intersection form Donaldson: definite intersection forms of smooth 4-manifolds must be diagonalizable - D_{IV}^5 has intersection form on H^2 determined by its topology - Question: does the BST integer structure force the intersection form to be standard? - The rank of H^2 of compact quotients of D_{IV}^5 — is it related to BST integers?

Deliverables

Toy C1 (Elie): $SU(2)$ instanton moduli on D_{IV}^5 slices. - $SU(2)$ embedding chain: $SU(2) \rightarrow SO(4) \rightarrow SO(5) \rightarrow SO(5,2)$ - Dimension of embedding: $\dim SU(2) = N_c = 3$, $\text{codim in } SO(5) = n_{\text{Crank}} - N_c = 7$ - Instanton number $k = 1$: $\dim M_{\{ASD\}}(S^4) = 81 - 3 = 5 = n_C - k = 2$: $\dim = 82 - 3 = 13 = 2g - 1$? - Do instanton moduli dimensions hit BST integers? - The “BPST instanton” on S^4 has $5 = n_C$ parameters (center + scale) - Target: 20-25 tests

Toy C2 (Lyra): Intersection form and BST integers. - For compact quotients ΓD_{IV}^5 : - Compute Betti numbers b_k from representation theory - $b_2 = \dim H^2$ — what is it for standard arithmetic quotients? - Signature = $b^+ - b^-$ — related to BST? - The Hirzebruch signature theorem: signature = L-polynomial evaluated on Pontryagin classes - For D_{IV}^5 : p_1 = first Pontryagin class, p_2 = second - Question: p_1 and p_2 of D_{IV}^5 in terms of BST integers - Target: 20-25 tests

Toy C3 (Lyra + Elie): Exotic R^4 exclusion via Bergman. - Cal’s tangent-space argument made computational: - The Bergman metric on $T_p(D_{IV}^5) = C^5$ induces a standard smooth structure on any R^4 slice - An exotic R^4 would require a non-standard smooth structure incompatible with the Hermitian inner product - Formalize: Kahler $\rightarrow$ standard smooth structure on all even-dim real submanifolds - Check: is this a known theorem? (Likely related to Newlander-Nirenberg) - The key claim: “BST physics lives on the standard R^4 because it inherits smoothness from D_{IV}^5 ” - Target: 15-20 tests

Paper potential: “Dimension 4 from Above: D_{IV}^5 and the Smooth Poincare Conjecture” — showing BST constrains the smooth category via its complex structure, explaining $d=4$ openness as under-determination. Venue: Geometry and Topology or Advances in Mathematics.

Honest scope: We will NOT resolve smooth Poincare. We aim to: 1. Show BST integers appear in Donaldson/SW invariant dimensions 2. Formalize why D_{IV}^5 forces standard R^4 smooth structure 3. Quantify the rank-2 excess as a connection to gauge theory moduli

Work Assignment Summary

Toy	Investigation	Owner	Tests	Dependencies
A1	ABC: Heegner Szpiro table	Elie	~25	None
A2	ABC: Non-CM Szpiro survey	Elie	~18	None
A3	ABC: Frey curve connection	Elie + Lyra	~18	A1
B1	Hilbert: Stark unit landscape	Lyra	~25	None
B2	Hilbert: Hecke L on D_{IV}^5	Lyra + Elie	~18	B1
B3	Hilbert: 9 Heegner unified	Lyra	~22	A1, B1
C1	Poincare: $SU(2)$ instanton moduli	Elie	~22	None
C2	Poincare: Intersection form	Lyra	~22	None
C3	Poincare: Exotic R^4 exclusion	Lyra + Elie	~18	C1, C2

Parallel tracks: A1+B1+C1+C2 can all start simultaneously (no dependencies). Second wave: A2+A3+B2+B3 after first results are in. Final: C3 after C1+C2.

Total: 9 toys, ~188 tests estimated. Three potential papers.

Success Criteria

Investigation	“Worth a paper” if...	“Document the boundary” if...
ABC	$\sigma = 3/2$ for all 6 Heegner + non-CM curves cluster near BST values	Only 49a1 is clean, others scatter
Hilbert 12th	Stark units for $Q(\sqrt{5})$ or $Q(\sqrt{7})$ are BST-structured	Real quadratic shows no BST pattern
Smooth Poincare	Instanton moduli dims are BST integers + exotic R^4 exclusion formalizes	Only counting identities, no new connection

In all cases: the boundary IS the result. “BST reaches this far and no further” is publishable if stated honestly.